INSTITUTE OF DISTANCE LEARNING

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

AE 351

Soil and Water Engineering I

Lecture 5

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HYDROLOGICAL FORECASTING AND DESIGN CRITERIA FOR HYDROLOGICAL STRUCTURES

Hydrological Forecasting

- A good hydrological forecast requires adequate records. Hydrological events are essentially random events with a non-cyclical nature.

e.g. a 100-year flood (that is a flood that will occur on average, once in every 100 years):

- May occur next year
- Or may not occur for 200 y
- Or may be exceeded several times in the next 100 years.

Accuracy and Duration for Data Analysis

- The accuracy of prediction depends how long the record is and for flood flows, one is lucky to have records longer than 30 y.

- Duration for data analysis may be based on water years or calendar years.

- A water year is measured between seasons of minimum flow.
- Calendar year is the normal calendar from January to December of each year.

Return Period

- **Recurrence interval or return period**, $\mathbf{T_r}$**:** Is the time that on average elapses between two events that equal or exceed a particular level **or** average interval in years within which a given event will be equalled or exceeded.

- The **N-year** event: The event that is expected to be equalled or exceeded, on average, every **N** years and has a recurrence interval $\mathbf{T_r}$ of **N** years.

Determining the Magnitude of the N-Year Event by Plotting

- Probability analysis seeks to define an event e.g. flood flow with a probability of being equalled or exceeded in any year. Return period, $\mathbf{T_r}$ is often used instead of probability to describe say a design flood.
- To plot a series of flows as a cumulative frequency curve, it is necessary to decide on a probability or return period to associate with each peak, known as the plotting position. Having listed a series of events (e.g. maximum floods) they are then each accorded a ranking **m**. Starting with:
- **m=1** for the highest value
- **m=2** for the next highest
- … And so on in descending order of magnitude.

Calculating Return Period

- The recurrence interval, $\mathbf{T_r}$ can be computed from one of a number of formulae:
  - **P=1/ T_r** i.e. Probability and return periods are reciprocals.
  - The probability P of an N-year event of return period $\mathbf{T_r}$ is: **P= 100%/T_r**

  - Weibull: **Tr=(n+1)/m** or **P=m/(n+1)** where **m**= event ranking
  - **n**=number of events

  - Gringorten: **T_r =(n+0.12)/(m+0.44)**
  - Hazens: **T_r =2n/(2m-1)**
  - Gumbel: **T_r =n/(m+c-1)** where **c** is Gumbel's correction factor

Gumbel's Correction Factors

m/n	1	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1	0.08	0.04
c	1	0.98	0.88	0.845	0.78	0.73	0.66	0.59	0.52	0.4	0.38	0.28

General Probability Equations

General form of exceedence probability $P(X \geq x_m) = \frac{m-b}{n+1-2b}$			
$P(X \geq x_m) = \frac{m}{n}$	the California's formula		
$P(X \geq x_m) = \frac{m-1}{n}$			
$P(X \geq x_m) = \frac{m-0.5}{n}$	Hazen formula	b=0.5	
$P(X \geq x_m) = \frac{m-0.3}{n+0.4}$	Chegodayev formula	b=0.3	
$P(X \geq x_m) = \frac{m}{n+1}$	Weibull formula	b=0	
$P(X \geq x_m) = \frac{m-0.375}{n+0.25}$	Blom formula	b=0.375	Normal distribution
$P(X \geq x_m) = \frac{m-0.44}{n+0.12}$	Gringorten formula	b=0.44	Extreme value type I

Intensity-Duration-Frequency Analysis of Rainfall Data

Irrigation and drainage systems are expected to serve for 5-10y for agricultural catchments, thus 30-40 years of data is required. For large catchments, 100-200 years of data may be necessary. Follow the following step by step procedure to do the analysis for the California method:

- Collect data for as many years of rainfall data as possible in the form of automatic rainfall recorder charts so that intensities can be calculated
- Study all significant storms and mark out the storms for 5, 10, 15, 20, 30, 45, 60 minutes duration or more for larger areas
- Tabulate the highest quantities of rainfall for all the durations for all the stations
- Arrange these highest rainfall values for the specified durations and calculate their intensities and arrange them in descending order of magnitude
- Assign rank numbers from the highest to the lowest i.e. 1,2,3,…,n
- Compute the recurrence interval or frequency of occurrence by taking the reciprocal of the probability
- Plot the intensities of rainfall against their durations for all the frequencies
- The plotting will lead to a series of parallel or nearly parallel curves, each representing the intensity-duration relation for the given frequency

Please see pages 78-79 of the lecture notes for an example.

DESIGN CRITERIA FOR HYDROLOGY

- If failure to a structure occurred in terms of the designed probability, then damage will ensue and one would have to decide if it is possible to exceed the design specification without structural failure. Use the formula or table to work out P based on policy decisions of the company.

- $P = 1-(1-1/T_r)\ L$

Where,

- L is an arbitrarily chosen life to give the probability of encountering the event, or something greater in that design life
- $\mathbf{T_r}$ is the return period of an event and
- P is the percentage probability.

PERCENTAGE RISK (P) OF ENCOUNTERING AN EVENT WITHIN A PARTICULAR DESIGN LIFE (L) FOR DIFFERENT RETURN PERIODS (T_r)

Design life, L (Years)	Return Period, T_r (Years)								
	5	**10**	**20**	**30**	**50**	**100**	**200**	**500**	**1000**
1	20	10	5	3	2	1	-	-	-
2	36	19	10	7	4	2	1	-	-
3	49	27	14	10	6	3	1	-	-
5	67	41	23	16	10	5	2	1	-
7	79	52	30	21	13	7	3	1	1
10	89	65	40	29	18	10	5	2	1
15	96	79	54	40	26	14	7	3	1
20	99	88	64	49	33	18	10	4	2
30	-	96	78	64	45	26	14	6	3
50	-	99	92	82	64	39	22	9	5
75	-	-	98	92	78	53	31	14	7
100	-	-	99	97	87	63	39	18	10
150	-	-	-	99	95	78	53	26	14
200	-	-	-	-	98	87	63	33	18
300	-	-	-	-	-	95	78	45	26
500	-	-	-	-	-	99	92	63	39
1000	-	-	-	-	-	-	99	86	63

Thank You

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